CZ CELLxGENE Discover is an online analytical platform and the largest repository of standardized single-cell data

CZI Single-Cell Team[a], Lattice Team[b], Clever Canary Team[c].
a) Chan Zuckerberg Initiative, Redwood City, CA, USA. b) Stanford University, CA, USA. c) Clever Canary, Santa Cruz, CA, USA.

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#CZCellxGene

A portal for the exploration of single-cell data

CZ CELL × GENE DISCOVER

CZ CELLxGENE Discover (cellxgene.cziscience.com) is a **free-to-use** online data portal hosting a growing corpus of single-cell data from modalities that include gene expression and spatial transcriptomics.

CZ CELLxGENE Discover provides a suite of tools that help researchers discover, download, and analyze single cell datasets.

UNIQUE CELLS
85M
Human 50M
Mouse 30M

DATASETS
1904

DISEASES
91

CELL TYPES
910

TOP TISSUES
Heart 1.7M
Blood 8.8M
Brain 14.4M
Lung 3.1M
Kidney 4M
Intestine 1.6M
Breast 1.7M

ASSAYS
- 10X seq
- sci-RNA-seq
- microwell-seq
- Drop-seq
- Seq-well
- microwell-seq
- Drop-seq
- Smart-seq2 and V4
- BD Rhapsody
- GEXSCOPE
- Visium 10X

GENE EXPRESSION
Normalized data
Raw counts

FEATURE METADATA
Name Type ID
empowered

SCHEMA

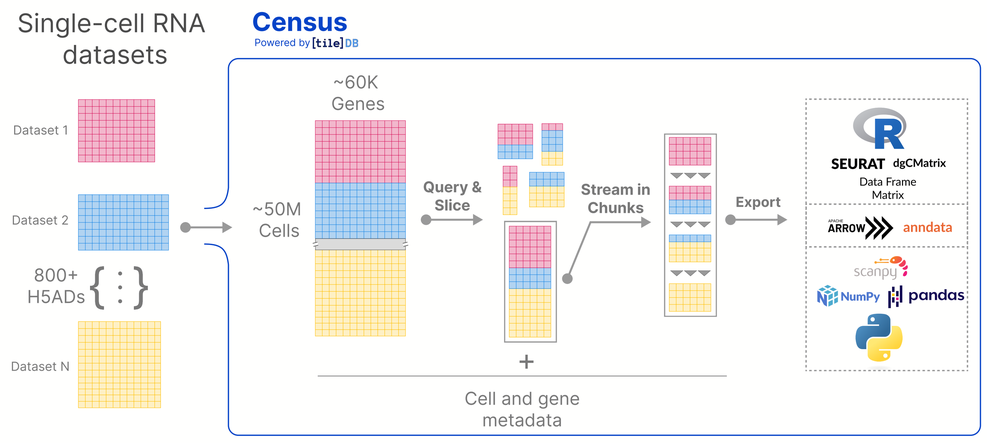
UNIVERSALLY AVAILABLE CELL METADATA
Species Tissue Age Sex Ethnicity
Assay Cell type Suspension Donor Disease

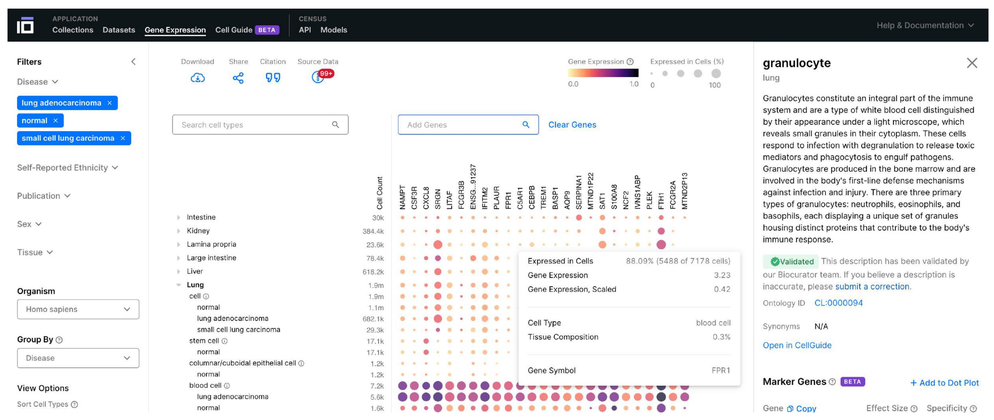
Integration made easier

All data in CZ CELLxGENE Discover are standardized to include **raw** and **normalized** counts, and annotated using an **ontological shared vocabulary** for cell and gene metadata.

Use Census for efficient access from Python and R

Census provides efficient access to all of our single-cell RNA data. Using **cell-based slicing and querying**, you can interact with the data across datasets through TileDB-SOMA, and **get slices in AnnData or Seurat objects.**



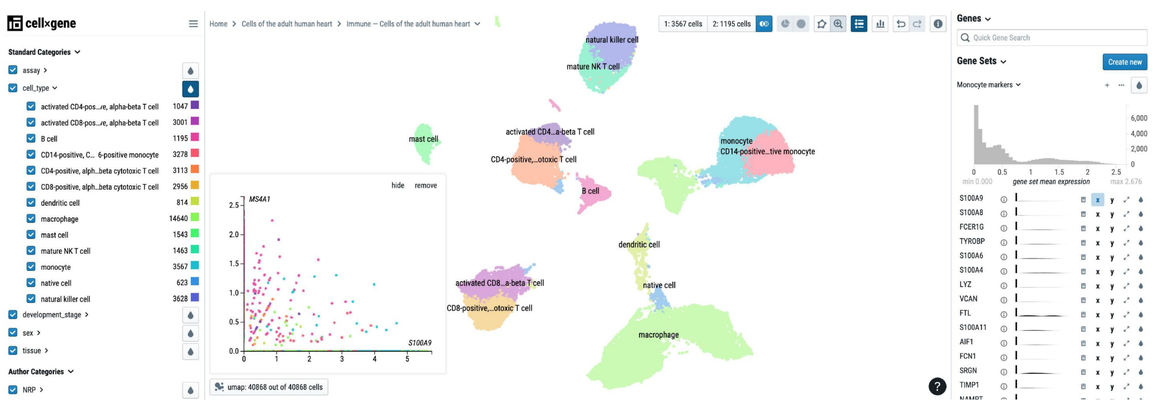


Explore gene expression on integrated data

Use Gene Expression to leverage **all data in the portal** and to query the expression of any gene across human and mouse cell types. Dot plots of gene expression are interactively created for any gene or tissue of interest.

Analyze individual datasets

On a 2-dimensional representation such as UMAP, t-SNE, or spatial coordinates, the CELLxGENE explorer allows **coloring cells** by metadata (e.g. cell type, disease) or gene activity, **subsetting** and analyzing subgroups of cells, performing **differential gene expression** and creating scatter plots of gene expression.



Partners

HUMAN CELL ATLAS
Tabula Sapiens
Tabula Muris
LungMAP
HTAN HUMAN TUMOR ATLAS NETWORK
KPMP
BICCN

Contribute and feedback

CZ CELLxGENE welcomes community data submissions. All data is hosted free of charge.

cellxgene@chanzuckerberg.com